# ARPIT KUMAR

*ML Engineer & AI Researcher | IIT Kharagpur | NLP · LLM/RAG · MLOps · Production ML*

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## PROFESSIONAL SUMMARY

Fourth-year dual degree student at IIT Kharagpur (Chemical Engineering + AI) with end-to-end Machine Learning (ML) experience spanning Natural Language Processing (NLP), Computer Vision (CV), LLM/RAG engineering, and MLOps. Ranked Top 0.5% globally in Amazon ML Challenge 2025 with a multimodal price prediction system full reproducibility via SHA-256 model fingerprinting, DVC, and MLflow. Developed state-of-the-art Wav2Vec2/WavLM fusion models for Speech AI research and deployed a RAG-based LLM support platform cutting resolution time by 35%. Codeforces Expert (peak 1612), AIR 135 in Integral Cup 2026, & Gold Medalist in IIT KGP's GC Data Analytics 2025. Seeking a Data Science internship to deliver full-stack AI/ML solutions with real product impact

## EDUCATION

**Indian Institute of Technology (IIT) Kharagpur** | Kharagpur, India — Jul 2022 - Apr 2027
*Integrated B.Tech - M.Tech (Dual Degree) in Chemical Engineering* — *GPA : 8.86/10*
*Micro-Specialization: Artificial Intelligence and Applications*

**Sri Sai Inter College, Barabanki** | Barabanki, UP — Jul 2020 - Jun 2022
*Intermediate | Science Stream | Uttar Pradesh Madhyamik Shiksha Parishad* — *Score : 85/100*

## WORK EXPERIENCE

**AI Researcher** | Internship — May 2025 - Jul 2025
*Prof. Shyamal Kumar Das Mandal | Advanced Technology Development Centre | IIT Kharagpur* — *On-site • Kharagpur, West Bengal, IN*

**Native Language Identification (NLI) from L2 English Speech**

- Engineered a speaker-balanced preprocessing pipeline for the NISP dataset, ensuring model fairness by standardizing 40/10/7 splits across 57 speakers per language to mitigate class imbalance and audio duration variability across 5 native Indian languages (Hindi, Telugu, Tamil, Kannada, and Malayalam)
- Developed and benchmarked seven deep learning architectures from CNN and ResNet-18/50 baselines to advanced fusion models by extracting normalized acoustic features like 128-bin Log-Mel spectrograms, 40-bin MFCCs with $\Delta/\Delta$-$\Delta$ coefficients to capture nuanced phonetic influence in L2 English speech
- Achieved a state-of-the-art validation accuracy of 0.88 using a Wav2Vec2 and WavLM-BiLSTM fusion model with frozen transformer layers, proving that context-aware self-supervised representations provide significant stronger discriminative cues for L1 language identification than classical spectral features
- Analyzed speaker separability and sociolinguistic variance by high-dimensional UMAP visualizations of embeddings, identifying overlaps between Hindi, Tamil pronunciation while quantifying tight, well-separated clusters for Telugu and Malayalam speakers for deep insights into regional language influence

***Technologies / Skills Used :*** *CNN (ResNet), BiLSTM, Mel-Spectrograms, MFCCs, t-SNE, Wav2Vec2, WavLM*

## PROJECTS

**Multimodal ML Platform Pipeline for E-Commerce** | Github Live Demo Website — Sep 2025 - Dec 2025
*AI Engineer | Top 0.5%, Amazon ML Challenge'25 — national competition (sponsored by Amazon)*

- Built a multimodal price prediction platform achieving 25.45 SMAPE (RMSE 0.82, MAE 0.66), as measured on validation benchmarks, by fusing SBERT text embeddings, CLIP image features, and engineered tabular inputs with a 5-fold stacked ensemble (RF, XGBoost, LightGBM, CatBoost + Ridge)
- Eliminated training-serving skew and ensured deterministic reproducibility, as measured by consistent reload outputs across environments, by implementing immutable run-scoped model bundles with SHA-256 fingerprinting, packaging vectorizers, feature selectors, dimensionality reducers, and model weights
- Architected production FastAPI serving layer featuring request-ID middleware, API-key protection, data-quality reject/quarantine policies, operational endpoints (/healthz, /readyz, /v1/predict), achieved 469.1 ms p95 end-to-end latency on 100 row smoke inference requests (2.04-39.09 ms/row observed)
- Improved experiment tracking and deployment reliability, as measured by auditable lineage across 46+ runs and automated CI/CD workflows, by integrating DVC, MLflow, and DagsHub with GitHub Actions pipelines for gated promotion, registry updates, smoke testing, and production readiness validation

***Technologies / Tools Used :*** *PyTorch, BERT, CLIP, Docker, Python, FastAPI, MLflow*

**LLM-Powered Employee Support Platform with RAG and Scalable AI Infrastructure** | Github — Jan 2025 - Apr 2025
*AI Engineer | 1st place, GC OpenSoft'25 — inter-hall software competition, IIT Kharagpur (sponsored by Deloitte)*

- Developed an LLM-powered employee support platform that reduced average resolution time by 35%, as indicated by session analytics, through a RAG-based conversational pipeline utilizing embedding stores (FAISS/Chroma), prompt-engineered workflows, and context-aware multi-turn interaction handling
- Enhanced answer grounding and decreased hallucination rate by 40%, as assessed through human evaluation audits, by integrating vector search, retrieval pre-processing, and citation-based response generation, ensuring evidence-backed outputs and increased reliability across various employee support queries
- Minimized inference latency and boosted API throughput by 50%, as measured by tracing metrics, through the implementation of async FastAPI endpoints, non-blocking backend-to-LLM communication, and optimized model routing strategies for efficient real-time response handling in production environments
- Cut model cost per session by 30% while enhancing observability and analytics, by applying token optimization, context summarization, and adaptive model selection, alongside containerized deployment, CI/CD pipelines, and monitoring with OpenTelemetry, Prometheus, and Grafana dashboards

***Technologies / Tools Used :*** *FastAPI, Python, LangChain, OpenAI API, Groq, PyTorch, Docker, JWT*

## SKILLS

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| **Programming Languages :** | Python, C++, SQL, C, JavaScript |
| **Frameworks & Libraries :** | PyTorch, TensorFlow, Scikit-learn, FastAPI, Flask, JAX, Numpy , Pandas, React |
| **Data Science :** | Machine Learning, Statistics, Probability, A/B Testing, Deep learning, Computer Vision, Data Analysis |
| **Generative AI :** | Retrieval-Augmented Generation, Model Interpretability, Distributed Inference, Quantization |
| **Tools & Platforms :** | MLflow, DVC, Git, GitHub, Docker, Dagshub, CI/CD, GitHub Actions |

## POSITION OF RESPONSIBILITY

**Developers' Society, TSG, IIT Kharagpur** — Sep 2023 - Present
*Advisor | Machine Learning & Software Architecture* — *Kharagpur, West Bengal, IN*

**Public Policy and Governance Society, IIT Kharagpur** — May 2023 - Sep 2024
*Executive Member | Quantitative Policy Research* — *Kharagpur, West Bengal, IN*

## AWARDS & ACHIEVEMENTS

- **Competitive Programming**: Achieved Expert rating on Codeforces (Peak: 1612), advance proficiency in data structures and algorithmic problem-solving
- **Quant & ML Challenges**: Ranked Top 20 national in DTL Quant Challenge 2024 & Top 0.5% global Amazon ML Challenge 2025 out of 50K+ participant
- **Mathematics**: Secured AIR 135 in the Integral Cup 2026 in 3 tracks Probability Theory & Statistics, Linear Algebra & Optimization and Integral Analysis
- **GATE 2026**: Achieved AIR 807 in the GATE 2026 CH stream (conducted by IIT Guwahati), ranking among the top tier of national engineering graduates
- **JEE 2022**: Secured AIR 1478 in JEE Advanced 2022 and a 98.28 percentile in JEE Mains among 1.1M+ candidates, the most competitive exam in India
- **Institute Championships**: Awarded 1 Gold Medal (General Championship Data Analytics 2025), 2 Silver Medals (Open IIT Data Analytics 2024 & Open IIT Case Study 2024), and 1 Bronze Medal (General Championship ChemQuest 2025) at IIT Kharagpur